On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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Esin

1-

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \xrightarrow[R_4-(-1)R_1]{R_2-(-8)R_1,\ R_3-m.R_1} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha+8 & 5+8\alpha & -\gamma & 1+8\alpha \\ 2-m & \beta-m\alpha & 3+m\gamma & \alpha^2-m\alpha \\ 2 & 14+\alpha & -6-\gamma & 2+\alpha \end{bmatrix} \xrightarrow[R_4-k.R_2]{R_3-(-1)R_2}$$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha+8 & 5+8\alpha & -\gamma & 1+8\alpha \\ 10+3\alpha-m & \beta+5+(8-m)k & 3+(m-1)\gamma & \alpha^2+(8-m)\alpha+1 \\ 2-3\alpha k-8k & 14-5k+(1-8k)\alpha & -6+(k-1)\gamma & 2-k+(1-8k)\alpha \end{bmatrix}$$

$$\xleftarrow{R_4-\frac{R_3}{2}} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha+8 & 5+8\alpha & -\gamma & 1+8\alpha \\ 10+3\alpha-m & \beta+5+(8-m)\alpha & 3+(m-1)\gamma & \alpha^2+(8-m)\alpha+1 \\ \frac{\alpha^2+(10-16k-m)\alpha+5-2k}{2} & & & \end{bmatrix}$$

a) $3\alpha+8=0$ $\quad \boxed{\alpha=-\frac{8}{3}}$

$10+3\alpha-m=0$ $\quad \boxed{m=2}$

$\beta+5+(8-2).\left(-\frac{8}{3}\right)=0$ $\quad \boxed{\beta=11}$

$14-5k-\frac{8}{3}+\frac{6k}{3}=0$

$\frac{9k}{3}=-\frac{34}{3}$

$\boxed{k=-\frac{34}{9}}$

$-6-\frac{83}{9}\gamma-\frac{3}{2}-\frac{\gamma}{2}=0$

$\frac{215\gamma}{98}=-\frac{15}{2}$

$\boxed{\gamma=-\frac{147}{43}}$

2-

CamScanner ile tarandı

3- $u.u^T = I \Rightarrow H = I - 2I \Rightarrow H = -I$

a) $H.H = -I.\overset{-I}{H} \Rightarrow H^2 = I^2 \Rightarrow \boxed{H^2 = I}$

b) —

c) $I \rightarrow$ because of $H = -I$

d) H is diagonalizable, because it is an identity matrix (I).

4- a) $\boxed{\begin{bmatrix} v_L H \\ w_L H \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_L \\ w_L \end{bmatrix}}$

b)

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